

Tanay Desai

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Education

Rutgers University–New Brunswick

Bachelor of Science in Computer Science & Minor in Data Science

Sep 2023 – May 2027

Shri Bhagubhai Mafatlal Polytechnic

Diploma in Information Technology

Dec 2020 – Jun 2023

Experience

Rutgers Office of Information Technology - Software Engineer

Feb 2024 – present

React.js, Redux, TypeScript, JavaScript

New Brunswick, NJ

- Collaborate with cross-functional developers to modernize and optimize Rutgers University's central web portal, an enterprise-scale platform supporting academic and student services, relied upon daily by over **70,000** students, faculty, and staff.
- Develop reusable and responsive React.js components, reducing development time by **30%** while enhancing overall code maintainability, scalability, and cross-project performance, enabling faster feature delivery and long-term application stability.
- Resolve critical software issues, improving overall system performance by **15%** and providing a smoother user experience.
- Engineered dark mode features, increasing user engagement by **20%** while improving visual appeal and overall experience.

Vertex Inc. - Software Engineering Intern (Backend)

Jun 2026 – Aug 2026

Java, Spring Boot, Docker, React.js

King Of Prussia, PA

- Remediated **18** security vulnerabilities across backend Java/Maven services, including a critical improper-authentication flaw (Snyk priority 655/1000) that exposed paths for remote code execution and sensitive-data access.
- Root-caused and fixed a data-integrity defect where soft-deleted certificates were incorrectly re-synced to O Series — including a subtle approval-workflow bug that silently flipped the delete flag from true to false — hardening sync-eligibility logic to exclude deleted records across all edits and state transitions.
- Audited **50+** LaunchDarkly feature flags across Certificate Center via the LD REST API, driving cleanup of stale flags and their dead code paths to reduce config sprawl and production risk.

Skillsvista - Front-End Engineering Intern

Jul 2022 – Aug 2022

Vue.js, Vuex, JavaScript

Dublin, Ireland

- Designed and developed **3+** responsive and reusable web components using Vue.js, improving user experience and overall performance for **2,000+** monthly users while ensuring seamless functionality across devices and platforms.
- Leveraged **Vuex** for robust and scalable state management, improving data flow, system stability, and overall application performance by ensuring consistent state synchronization across components and reducing potential errors.
- Diagnosed and resolved **5+** critical software bugs across core modules, reducing support tickets by **15%** while enhancing overall system stability, minimizing recurring errors, and improving the end-user experience.

Projects

SignalWatch | *TypeScript, Next.js, React, PostgreSQL, Prisma, NextAuth, Tailwind*

- Built an AI-powered stock monitoring pipeline that ingests real-time BSE/NSE corporate announcements and uses Claude (Anthropic SDK) to auto-summarize, categorize into 10 types, score importance (1–10), and tag sentiment; then fans out preference-based email alerts and live updates via Server-Sent Events.
- Designed and shipped a full-stack Next.js 16 / React 19 / TypeScript app (deployed on Vercel) with NextAuth authentication, a normalized PostgreSQL/Prisma schema (11 models spanning watchlists, notification preferences, and monitoring logs), CSV portfolio import, and a background job system with per-source delivery logging.

Epoch-OS | *Rust, TypeScript, RAG, Docker, React, LLMs (Ollama/OpenAI), Tauri, ChromaDB, CI/CD*

- Directed the frontend architecture and implementation using React, TypeScript, and Tauri to deliver a seamless, high-performance, and scalable cross-platform user experience, improving load times by **20%** and reducing UI-related defects by **15%**.
- Led the development of an intelligent operating environment that integrated AI agents and advanced Voice AI capabilities into digital workflows and productivity tools, automating repetitive tasks and improving overall efficiency by nearly **30%**.
- Built a Retrieval-Augmented Generation (RAG) pipeline leveraging ChromaDB and large language models (LLMs) to deliver highly domain-specific and context-aware AI responses, increasing query accuracy by nearly **35%** and reducing response latency by **20%**.

Technical Skills

Programming Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, R, C#, Rust

Libraries and Frameworks: React.js, Node.js, Next.js, Vue.js, Express.js, Flask, .NET, Tauri, Spring Boot

Databases: MongoDB, MySQL, Oracle, PostgreSQL, SQL, SQLite

Tools & Platforms: Git, GitHub, Firebase, Postman, VS Code, Docker, ChromaDB

Achievements

- Orientation Leader for **6000+** students; awarded **Best Orientation Leader of the Session** for leadership excellence.
- President of PolyHacks, a flagship hackathon with **200+** participants; managed planning, sponsorships, and team coordination.
- Secured **Second Place** in a Competitive Programming Competition, showcasing expertise in algorithms and coding efficiency.
- Elected **Student Representative** for Diploma; championed student interests and strengthened communication with faculty.